Q1. What is Accuracy?

Accuracy refers to how close a measured or calculated value is to the true (actual) value. If the obtained result is very near to the true value, it is said to be accurate. In numerical methods, accuracy indicates the correctness and reliability of a result. A highly accurate result has a very small difference from the true value, while a less accurate result has a larger difference. Thus, accuracy helps in judging how correct a computed value is.

Example

If the true value is 100 and the calculated value is 98, then the result is very close to the true value, so it is considered accurate. If the calculated value is 80, then it is far from the true value, so it is less accurate.

Types of Accuracy

1. High Accuracy:
When the calculated value is very close to the true value, it is called high accuracy. In this case, the error between the true value and approximate value is very small or negligible, making the result reliable.
Example: True value = 100, Approximate value = 99 → Very high accuracy

2. Low Accuracy:
When the calculated value is far from the true value, it is called low accuracy. In this case, the error is large, which makes the result less reliable and less correct.
Example: True value = 100, Approximate value = 80 → Low accuracy

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Q2. What is Error?

Error is the difference between the true (actual) value and the approximate (measured or calculated) value. It represents how much the obtained result deviates from the correct value. In numerical methods, error is used to measure the accuracy of a result. If the error is small, the result is considered accurate, and if the error is large, the result is considered less accurate. Thus, error helps in determining the reliability of calculations.

Example

If the true value is 100 and the calculated value is 95, then
Error = 100 − 95 = 5

Another case:
If the calculated value is 105, then
Error = 100 − 105 = −5

Types of Error

1. Positive Error:
When the approximate value is less than the true value, the error is called positive error. In this case, the calculated result is smaller than the actual value, so the difference comes out positive. This type of error shows that the value is underestimated.
Example: True value = 100, Approximate value = 95 → Error = +5

2. Negative Error:
When the approximate value is greater than the true value, the error is called negative error. In this case, the calculated result is larger than the actual value, so the difference becomes negative. This type of error shows that the value is overestimated.
Example: True value = 100, Approximate value = 105 → Error = −5

Q3. What are Significant Digits?

Significant digits are the digits in a number that carry meaningful information about its accuracy. These include all certain digits and the first uncertain (estimated) digit in a measurement. Significant digits indicate the precision of a value, meaning how accurately the number is known. More significant digits mean higher accuracy, while fewer significant digits mean lower accuracy.

Example

In the number **345.67**, all digits are significant, so it has 5 significant digits. In **0.0045**, only **4 and 5** are significant, so it has 2 significant digits (leading zeros are not counted).

Types / Rules of Significant Digits (20%)

1. Non-zero digits:
All non-zero digits (1–9) are always significant because they represent actual measured values.
Example: 456 → 3 significant digits

2. Zeros between non-zero digits:
Zeros placed between non-zero digits are significant because they lie between meaningful digits.
Example: 1002 → 4 significant digits

3. Leading zeros:
Zeros at the beginning of a number are not significant because they only show position of decimal point.
Example: 0.0056 → 2 significant digits

4. Trailing zeros:
Zeros at the end of a number may or may not be significant depending on the decimal point. If a decimal point is present, they are significant.
Example: 2.300 → 4 significant digits

Q4. What are Rounding Off Digits?

Rounding off digits is the process of simplifying a number by reducing the number of digits while keeping its value close to the original number. It is done to make calculations easier and results more manageable. In this method, digits are removed according to certain rules, and the remaining number is adjusted to maintain accuracy. Rounding off helps in representing numbers in a simpler form without causing much error.

Example

Round **3.4567** to 3 decimal places → 3.457
Round **8.342** to 2 decimal places → 8.34

Types / Rules of Rounding Off

1. If the digit to be dropped is less than 5:
If the digit next to the last retained digit is less than 5, the number remains unchanged.
Example: 4.732 → 4.73 (2 decimal places)

2. If the digit to be dropped is greater than 5:
If the digit is greater than 5, the last retained digit is increased by 1.
Example: 6.278 → 6.28 (2 decimal places)

3. If the digit to be dropped is exactly 5:
If the digit is 5, the last retained digit is increased by 1 (standard rule used in exams).
Example: 2.345 → 2.35 (2 decimal places)

Q5. What is Truncation?

Truncation is the process of reducing a number by simply cutting off digits beyond a certain point without rounding them. In this method, the unwanted digits are removed, and the remaining number is taken as it is. It makes calculations simpler but may introduce more error compared to rounding off because no adjustment is made. Truncation is commonly used in numerical methods where simplicity is preferred over exact accuracy.

Example

Truncate **3.4567** to 3 decimal places → 3.456
Truncate **8.349** to 2 decimal places → 8.34

Types of Truncations

1. Decimal Truncation:
In this type, digits after a fixed number of decimal places are removed without rounding. The value becomes shorter but slightly less accurate.
Example: 5.6789 → 5.67 (2 decimal places)

2. Integer Truncation:
In this type, the decimal part of a number is completely removed, leaving only the integer part. This causes a larger loss of accuracy.
Example: 9.87 → 9

Q6. Absolute Error

Absolute error is the difference between the true (actual) value and the approximate (measured or calculated) value, taken as a positive value. It shows the magnitude of error without considering its direction (sign). In numerical methods, absolute error is used to measure how far a result is from the true value. Smaller absolute error indicates higher accuracy, while larger absolute error indicates lower accuracy.

Formula: Absolute Error = |True Value − Approximate Value|

Example

If the true value is 100 and the approximate value is 95, then
Absolute Error = |100 − 95| = 5

If the approximate value is 105, then
Absolute Error = |100 − 105| = 5

Types of Absolute Error

1. Small Absolute Error:
When the difference between true value and approximate value is very small, it is called small absolute error. This indicates that the calculated result is very close to the true value and hence highly accurate.
Example: True value = 50, Approximate value = 49.8 → Error = 0.2

2. Large Absolute Error:
When the difference between true value and approximate value is large, it is called large absolute error. This indicates that the result is far from the true value and less accurate.
Example: True value = 50, Approximate value = 40 → Error = 10

Q7. Relative Error

Relative error is the ratio of absolute error to the true (actual) value. It shows the size of the error in comparison to the true value, giving a better idea of accuracy than absolute error. In numerical methods, relative error helps in understanding how significant the error is with respect to the actual value. A smaller relative error indicates higher accuracy, while a larger relative error indicates lower accuracy.

Formula: Relative Error = Absolute Error / True Value

Example

If the true value is 100 and the approximate value is 95, then
Absolute Error = |100 − 95| = 5
Relative Error = 5 / 100 = 0.05

Types of Relative Error

1. Small Relative Error:
When the ratio of absolute error to true value is very small, it is called small relative error. This means the error is negligible compared to the true value, and the result is highly accurate.
Example: Absolute Error = 1, True Value = 100 → Relative Error = 0.01

2. Large Relative Error:
When the ratio of absolute error to true value is large, it is called large relative error. This means the error is significant compared to the true value, and the result is less accurate.
Example: Absolute Error = 10, True Value = 20 → Relative Error = 0.5

Q8. Percentage Error

Percentage error is the relative error expressed in percentage form. It is obtained by multiplying the relative error by 100. It shows how large the error is compared to the true value in percentage terms, which makes it easier to understand and compare accuracy. In numerical methods, a smaller percentage error indicates higher accuracy, while a larger percentage error indicates lower accuracy.

Formula:
Percentage Error = (Absolute Error / True Value) × 100

Example

If the true value is 100 and the approximate value is 95, then
Absolute Error = |100 − 95| = 5
Percentage Error = (5 / 100) × 100 = 5%

Types of Percentage Error

1. Small Percentage Error:
When the percentage error is very low, it is called small percentage error. This means the calculated value is very close to the true value, and the result is highly accurate.
Example: Error = 2% → Result is highly accurate

2. Large Percentage Error:
When the percentage error is high, it is called large percentage error. This means the calculated value is far from the true value, and the result is less accurate.
Example: Error = 25% → Result is less accurate
